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Magnetostratigraphy of a Middle Jurassic delta system (Lajas Formation), Portada Covunco section, southern Neuquén Basin, Argentina

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ABSTRACT

A systematic paleomagnetic study was carried out in the Lajas Formation (Middle Jurassic), cropping out at the Covunco creek in the central part of the Neuquén basin. The Lajas Formation at the Portada Covunco section consists of a succession of conglomerates, sandstones and mudstones that reaches c. 290 m-thick. On the basis of lithological and textural features, sedimentary structures, strata geometries, and fossil content, 14 facies were defined that were grouped in two facies associations within a fluvial-dominated delta system. The age of the Lajas Formation is rather controversial, and has been assigned from sequence stratigraphy and palynological data. Thus from south to north, interpreted ages span the Upper Aalenian–Middle Bathonian through the Callovian. Therefore, magnetostratigraphy was used to better precise the age of the Formation at Portada Covunco. Paleomagnetic data obtained from 29 sampling sites present reliable behaviors. Rock magnetic studies (thermomagnetic curves, K vs H curves, IRM acquisition curves) suggest that the remanence is carried by Ti-poor magnetite. Anisotropy of magnetic susceptibility data show clustering of the K_{max} axes produced by the action of paleocurrents. The characteristic remanent magnetization passed a tilt test and a reversal test that prove its primary origin. The resultant magnetostratigraphy is characterized by a reverse-dominant with minor normal polarity intervals. When compared with the Bajocian-Bathonian reference geomagnetic polarity time scale (GTS2016), the best fit is achieved in the Lower through Middle Bathonian (Chrons M41 to M39). A paleomagnetic pole was calculated for Portada Covunco located at 167.1°E, 59.4°S, N = 28, A_{95} = 3.6°, K = 57.2. When compared with the other available paleopole of the same age −170 Ma-derived from dykes exposed at Mamil Choique, Patagonia, Portada Covunco falls in the same paleomeridian although at a lower latitude. Thus, it is interpreted that the studied section was not subjected to rotations around vertical axis (R = 4° ± 10°), but was compacted, which produced inclination shallowing of the magnetic vector (F = 18° ± 11°). The calculated inclination shallowing and amount of compaction in the section are well within the expected values in medium to fine-grained sandstones. Using the paleomagnetic pole from Mamil Choique as reference, the locality of Portada Covunco would have been located at c. 33°S during the Bathonian, in agreement with paleoclimatic interpretations derived from palynological data.

1. Introduction

Middle Jurassic lithostratigraphic units in the Neuquén basin constitute one of the most prolific conventional and non-conventional hydrocarbon systems in Argentina (Arregui et al., 2011; Jorgensen et al., 2013; Raggio et al., 2014). Such units compose the Cuyo Group where the deep marine mudstones of Los Molles Formation (Pliensbachian–lower Callovian) represent the source rock and unconventional shale reservoir, and the paralic sandstones of the Lajas Formation (upper Bajocian–lower Callovian) represent a tight reservoir. Since the early works of Gulisano (1981) and Gulisano et al. (1984), great progress has been made in the sedimentological and stratigraphic understanding of the Lajas Formation (Zavala, 1993; 1996a, 1996b; Spalletti, 1995; Zavala and González, 2001; McIlroy et al., 2005; McIlroy, 2007; Canale et al., 2015; Gugliotta et al., 2015; Kurcinka et al., 2017; Steel et al., 2018). Nonetheless, its age is still interpreted based on its stratigraphic

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position and sequence stratigraphic correlations with the Los Molles Formation, due to the lack of diagnostic fossils or radiometric dating.

The goal of this study is to refine the age of the Lajas Formation in the area of Portada Covunco based on magnetostratigraphic data. In the vicinity, comprehensive sequence stratigraphic correlations (Zavala, 1993; 1996a, 1996b; Zavala and González, 2001), assign a Late Bajocian-Middle Bathonian to the unit, although palynological studies indicate an Early Callovian age for the uppermost part (Martínez, 2002; Martínez et al., 2002). This is the reason why magnetostratigraphy can become an efficient tool to constrain ages. Magnetostratigraphic studies in the Jurassic of the Neuquén Basin includes the Lower Jurassic (e.g. Iglesia Llanos, 1997; 2012; Iglesia Llanos and Riccardi, 2000) and the Upper Jurassic - Lower Cretaceous (Iglesia Llanos et al., 2017; Kohan Martínez et al., 2018), so this represents the first magnetostratigraphic study in the Middle Jurassic from Argentina. For the Bajocian-Bathonian, the reference Greomagnetic Polarity Time Scale features a high rate or reversals of the Earth's magnetic field (Chrons M44 to M39), along with a difficult availability of ocean magnetic anomalies. Hence, the latest Geomagnetic Polarity Time Scale (GTS2016) includes two magnetostratigraphic scales for this time lapse, one made up from deep-tow magnetic data on Japanese sea-floor, and the other from sections on land calibrated with ammonites and nannofossils in the Sub-Mediterranean and Sub-Boreal realms. Both look slightly different, which urges the necessity of being cautious about the selection of the reference scale. According to Ogg and Hinov (2012), the results of direct modelling from deep-tow surveys for pre-M25r marine magnetic anomalies vary with the depth of the model, and remark that on-land magnetostratigraphies for the Oxfordian through Bajocian are consistent with an intermediate geophysical model. Therefore, we regarded the on-land composite scale from the Geomagnetic Polarity Time Scale 2016 as reference for the correlation with Portada Covunco.

Anisotropy of magnetic susceptibility (AMS) measurements showed a primary sedimentary fabric with clustering of the K_{max} axes and imbrication of the K_{min} axes. From comparison with paleocurrent data derived from the same unit in the area, AMS reflect the action of paleocurrents that aligned the fabric with the flow. Therefore, AMS proved to be a very effective tool for measuring paleocurrents for the first time, in rocks of the Lajas Formation where sedimentary structures are not conspicuous.

2. Geological setting

2.1. The Neuquén basin

The Neuquén basin covers a surface of more than 137,000 km^2 (Urien and Zambrano, 1994) in the provinces of Mendoza and Neuquén, western central Argentina (Fig. 1), and is the best studied in Argentina due to its relevance for the gas and oil industry. The basin is interpreted as a back-arc type, developed over continental crust by the end of the Triassic due to the thermo-tectonic collapse behind a stationary magmatic arc on the west (Mpodozis and Ramos 1989). The basin infill consists of a c. 7000 m-thick marine and continental sedimentary rocks that span the Late Triassic – Cenozoic. The first studies carried out on a regional scale, focused on stratigraphy, fossil content and structure (Weaver, 1931; Groeber, 1918, 1929, 1946; Herrero Ducloux, 1946; De Ferraris, 1947; Groeber et al. 1953).

The tectonic history of the basin is rather complex due to the occurrence of several major orogenies (Ramos, 1999). The first is the Gondwana Orogeny, an extensional back-arc regime that took place during the Carboniferous-Middle Permian, followed by the compressive San Rafael Orogeny in the Middle Permian (Ramos, 1988, Mpodozis and Ramos 1989). During this time, the collision of the Patagonia terrane with Gondwana would have prompted a regional penetrative deformation of the basement in the area of the Huincul Arch located to the south of the study area (Mosquera et al. 2011) (Fig. 2). By the Late Permian – Early Triassic, a generalized extensional tectonic regime initiated the onset of acidic volcanism and tectonic subsidence (Llambías et al., 2003, 2007; Leanza et al. 2005; Llambías and Leanza, 2005; Schiuma and Llambías, 2008). The Early Jurassic – Late Cretaceous was characterized by a thermal subsidence regime with localized tectonic events that led to the development of a shallow marine basin, connected to the proto-Pacific Ocean by narrow corridors across the volcanic island arc to the west (Legarreta and Uliana, 1996). Depocenters were filled by continental and marine siliciclastic, carbonate and evaporite successions (Cuyo, Lotena, and Mendoza Groups). During the middle - Late Cretaceous, the Andean Orogeny prompted a major compression caused by the subduction of the Pacific plate under the South American margin started the formation of the Agrio fold and thrust belt in the Paleogene (Zapata and Folguera, 2005). Finally, by the Middle Miocene the Quechua Orogeny developed in the Chilean-Argentinian Andes prompting the reactivation of the Agrio fold and thrust belt (Zapata and Folguera, 2005).

The area of Portada Covunco is part of the Agrio fold and thrust belt (Fig. 2) in the western sector of the basin. In Portada Covunco, the Cuyo Group comprises from base to top (Figs. 1 and 2), the Los Molles (Pliensbachian-Bathonian), Lajas (Aalenian-Callovian) and Tábanos (Callovian) Formations, followed by the Lotena Group (Callovian-Kimmeridgian). The overlaying the Mendoza Group spans the Kimmeridgian-Barremian capped by the Macho Viejo Basalt (Pleistocene).

2.2. Lajas Formation at Portada Covunco

The Lajas Formation (Weaver, 1931) is part of the regressive second order cycle of the Cuyo Group (Dellapé et al. 1978), also known as Cuyo Mesosequence (Legarreta and Gulisano, 1989) or Cuyo Subsynthem (Riccardi and Gulisano, 1992). In southern Neuquén, the base of the Cuyo Group (Early Pliensbachian) represents a regional marine transgression. It overlaps the deep marine deposits of the Los Molles Formation (Weaver, 1931), either on the shallow-marine limestones of the Chachil Formation (Leanza, 1992) or the continental facies of the Precuyo Group, where the Intraliassic (Gulisano et al., 1984) or Rioatuelic unconformity (Leanza, 2009) lay. This succession is followed by the deltaic, shallow-marine Lajas Formation (Early Bajocian to Early Callovian) and fluvial Challacó Formation (Bathonian-Early Callovian). The Lajas Formation is a well-known deltaic succession (Gulisano, 1981; Gulisano et al., 1984), that is interpreted as tide-dominated (McIlroy et al., 1999; Brandsaeter et al. 2005), wave dominated (Spalletti, 1995; Gugliotta et al. 2015) to fluvial dominated (Zavala and González, 2001; Canale et al., 2015; Kurcinka et al., 2017; Steel et al., 2018).

The Lajas Formation at Portada Covunco (38°48′22″S, 70°12′32″W), makes up the core of the Covunco anticline (Fig. 2), where it reaches up to 290 m-thick. In this area, Zavala and González (2001) divided the Lajas Formation in two sequences: Sequence 1 can reach up to 270 m-thick, and includes siltstones and fine to medium-grained sandstones, eventually interlaid with conglomerates. Sandstones are mainly composed by quartz and feldspar grains., Sequence 2 is made up of coarse-grained to conglomeratic sandstones with minor intercalations of siltstones. Composition of sandstones in this Sequence includes mostly lithic components. Paleocurrents measured in coarse-grained sandstones and conglomerates show provenance from the southwest.

The age of the Lajas Formation is diachronous from south to north. In the central and southern areas of the Neuquén basin, the unit is interpreted as Upper Aalenian–Middle Bathonian from sequence stratigraphy and palynology. In the northern part however, it is assigned to the Callovian based on ammonites (Westermann and Riccardi, 1972, Riccardi et al., 1991, Gulisano et al., 1984, Riccardi and Gulisano, 1992, Zavala, 1993, 1996a, 1996b García et al., 1994, Martínez et al., 2002, Garcia et al., 2006, Zavala and González, 2001, among others). The Portada Covunco section is interpreted Late Bajocian to Middle Bathonian based on sequence stratigraphy: Sequences 1 is assigned to the Late Bajocian and sequence 2 to the Early-Middle Bathonian (Zavala

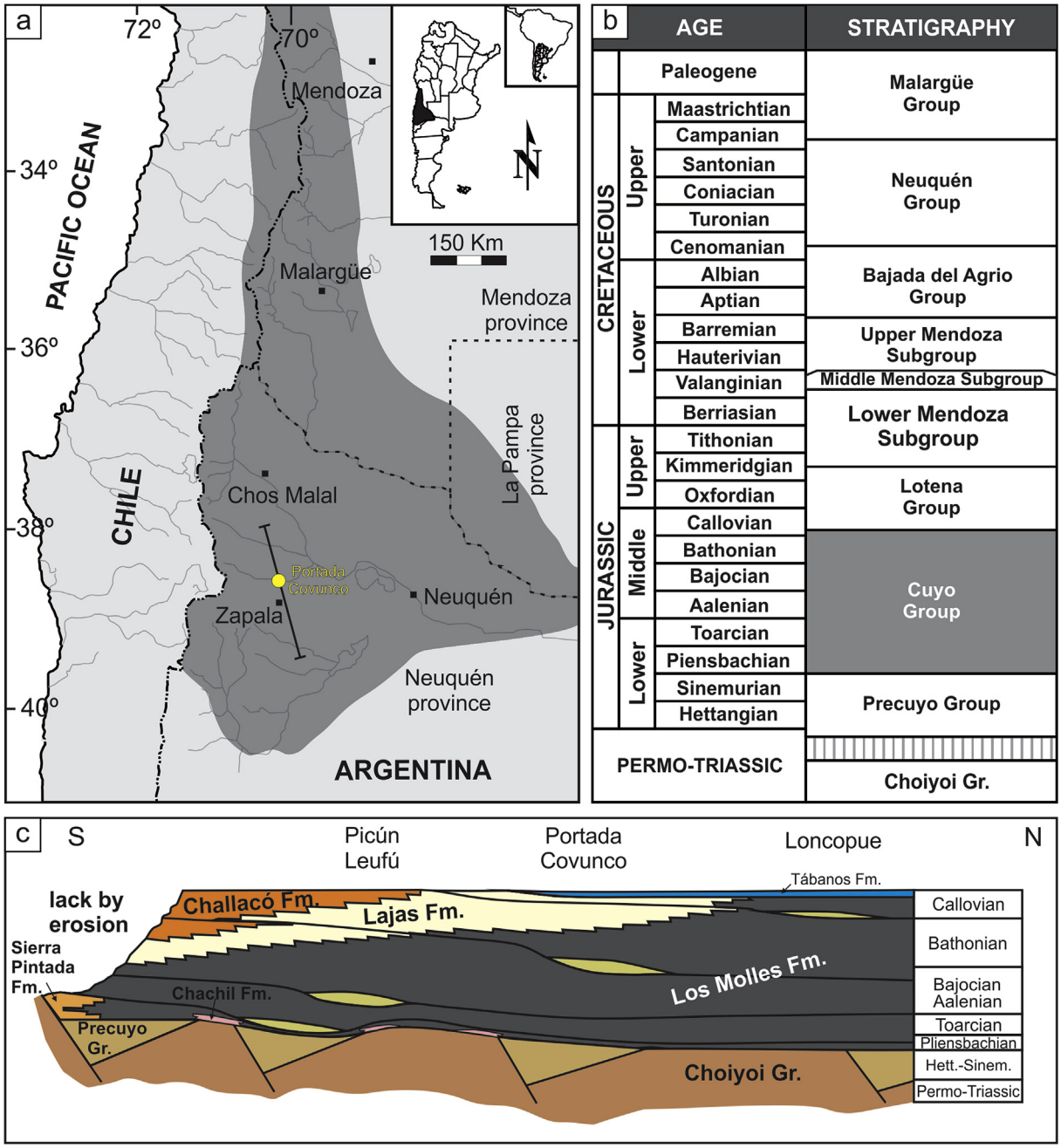


Fig. 1. a) Location map of the Neuquén basin at Portada Covunco. b) Main stratigraphic divisions of the Neuquén basin. c) Lithostratigraphic subdivision of the Cuyo Group (modified from Legarreta and Gulisano, 1989).

and González, 2001). Palynological data on the other hand, suggest an Early Callovian age for the uppermost part of Sequence 2 (Martínez, 2002; Martínez et al., 2002).

During paleomagnetic sampling, we found ammonites from the Late Bajocian *M. rotundum* Zone (Riccardi pers. comm. 2017) in the middle to upper part of Los Molles Formation at Arroyo Carreri (38°52′54″S, 70°26′36″W), a locality close to Portada Covunco (38°48′22″S, 70°12′32″W). Here, *ex situ* specimens from the Lajas Formation indicate an Early to Late Bathonian age for this unit (Riccardi, pers. comm. 2017). Assuming that the transition between the Los Molles and Lajas Formations occurs at a similar age in Portada Covunco, and taking into account the previous stratigraphic and paleontological information, the age of the Lajas Formation in the study section would be Upper Bajocian–Bathonian, reaching probably the Lower Callovian.

3. Sedimentology of the Lajas Formation at Portada Covunco section

The Lajas Formation at the Portada Covunco section is made up of conglomerates, sandstones and mudstones, reaching c. 290 m thick (Fig. 3a). The measured sedimentary section was described based on lithological and textural features, sedimentary structures, strata geometries, and fossil content (Table 1) that allow us to define 14 facies. We grouped them in two facies associations, which we interpret as a fluvial-dominated delta system.

3.1. Facies association 1: delta front

Description: Facies association 1 is dominant in the section (Fig. 3a). It consists of coarsening upward cycles with laminated mudstones (Fl),

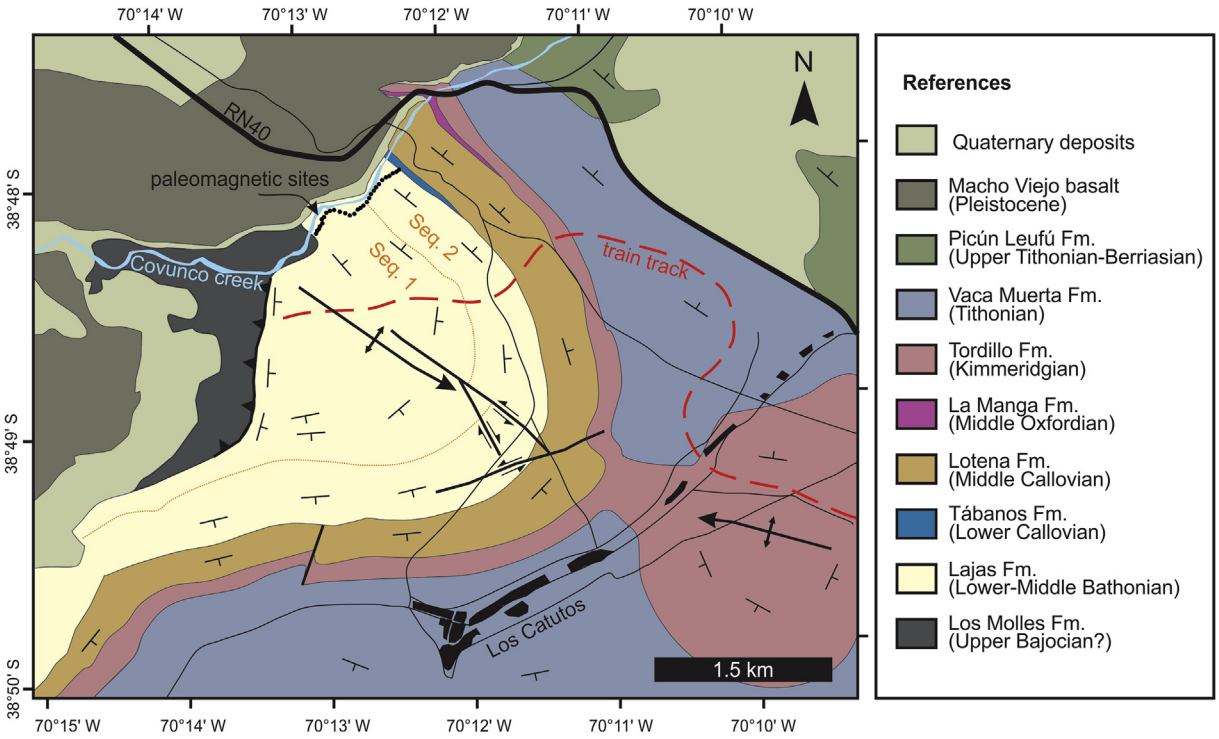


Fig. 2. Geological map of the Portada Covunco area. The sampled section is located at 38°48′ 22″S, 70°12′ 32″W. Black dots: sampling sites, from base (SW) to top (NE). Seq.1, Seq. 2: Sedimentary Sequences 1, 2.

horizontal laminated sandstones (Sh), ripple laminated sandstones (Sr), bioturbated massive sandstones (Sm), sandstones with convolute lamination or fluid-scape structures, that pass to planar or trough cross-stratified sandstones (Sp, St), and planar stratified coarse-grained sandstones to fine-grained conglomerates (SGp, Gp) (Fig. 3b–c). In the middle and upper parts of the section, the coarsening upward cycles present thin beds of massive or graded bioclastic oyster-dominated concentrations (Sbm, Sbg) showing erosional bases. Sandstones are lithoarenites, with a variable content of feldspars and volcanic lithic fragments. Paleocurrent directions in this facies association were measured on planar and trough cross-stratification. They have a mean direction of Az = 51° (n = 24) and range from 290° to 73° (Fig. 3d). Bioturbation is restricted to Sh, Sr, Sm and Sp facies. Trace fossil are dominated by *Ophiomorpha* isp. (Skolithos ichnofacies), as well as the *Paleophycus* isp., *Thalassinoides* isp., *Gordia* isp. and *Gyrochorte* isp. (Cruziana ichnofacies).

Interpretation: The facies association 1 is interpreted as a delta front dominated by fluvial action. The coarsening-upward cycles represented by large-scale cross-stratified sandstones are interpreted as constructive mouth bars (Van Heerden and Roberts, 1988; Plink-Björklund and Steel, 2005; Bhattacharya, 2006). Sandstones accumulated in the proximal part of the mouth bar become unstable at times of high sediment input, prompting gravitational deposits (Mulder et al., 1996). In distal mouth bars, the episodic intercalation of laminated mudstones (Fl), horizontal laminated sandstones (Sh), ripple laminated sandstones (Sr), and bioturbated massive sandstones (Sm) are probably related to hyperpycnal flows, which are not deposited in the mouth bars due to the inertia and floating force of sediment plumes (Bates 1953; Mulder et al. 1996). Convolute bedding is related to rapid deposition on water-saturated sediments, causing an expulsion of the fluids contained in pores (Oliveira et al., 2011). Massive to graded bioclastic concentrations (Sbm, Sbg) are interpreted as storm reworked oyster reefs, formed on distal levee ridges, during abandonment phase of delta-lobes (Elliott, 1986; Starek et al., 2010; Hampson et al., 2011). The mixture of elements of the ichnofacies indicates fluctuations in fluvial discharges and variations in the water energy, sedimentation rate and salinity (Pemberton et al., 1992; MacEachern et al., 2005; Canale et al., 2016).

3.2. Facies association 2: hyperpycnal prodelta lobes

Description: Facies association 2 consists of 50–80 cm thick thinning-upward strata, with erosional bases, massive or graded conglomerates (Gcm, Gcg) in the lower part, and cross-stratified or horizontal laminated sandstones (St, Sp or Sh) in the upper part (Fig. 3e). These strata are interbedded within 1–2 m-thick horizontal laminated mudstones (Fl) and heterolithic deposits (S_{het}), as well as coarsening-upward horizontal to ripple laminated sandstones. Paleocurrent directions were measured on sandstones with cross-stratification. They have a mean direction of 85° (n = 10) and range from 60 to 108° (Fig. 3e). Traces fossil in this facies association are less abundant and diverse than in facies association 1, they are present only in the Sp facies and correspond to *Ophiomorpha* isp. and *Thalassinoides* isp.

Interpretation: The facies association 2 represent hyperpycnal-lobes in a prodelta setting (Canale et al., 2015). The presence of decimeter to meter-scale thinning-upward massive and graded conglomerate to sandy strata are interpreted as channel-lobes originated by deaccelerating hyperpycnal flows (Zavala et al., 2006). Horizontal laminated mudstones indicate deposition in relatively quieter environment. Heterolithic intervals indicate high aggradation rates probably triggered by rapid flow deceleration, consistent with hyperpycnal flows incursion into standing or low energy water. Overall, the association of channels and inter-channel deposits with marine trace fossils are interpreted to record a prodelta setting, hence representing an environment of deposition slightly basinward of facies association 1, the delta front.

4. Experimental methodology

A total of 29 sampling horizons or sites were sampled for paleomagnetic analyses at Portada Covunco (Fig. 3). In each site, at least 5 cores were drilled in medium to fine-grained sandstones using a portable gasoline-powered drill that were oriented with sun and Brunton compasses. Cores were sliced in 280 standard −2.2 cm-specimens.

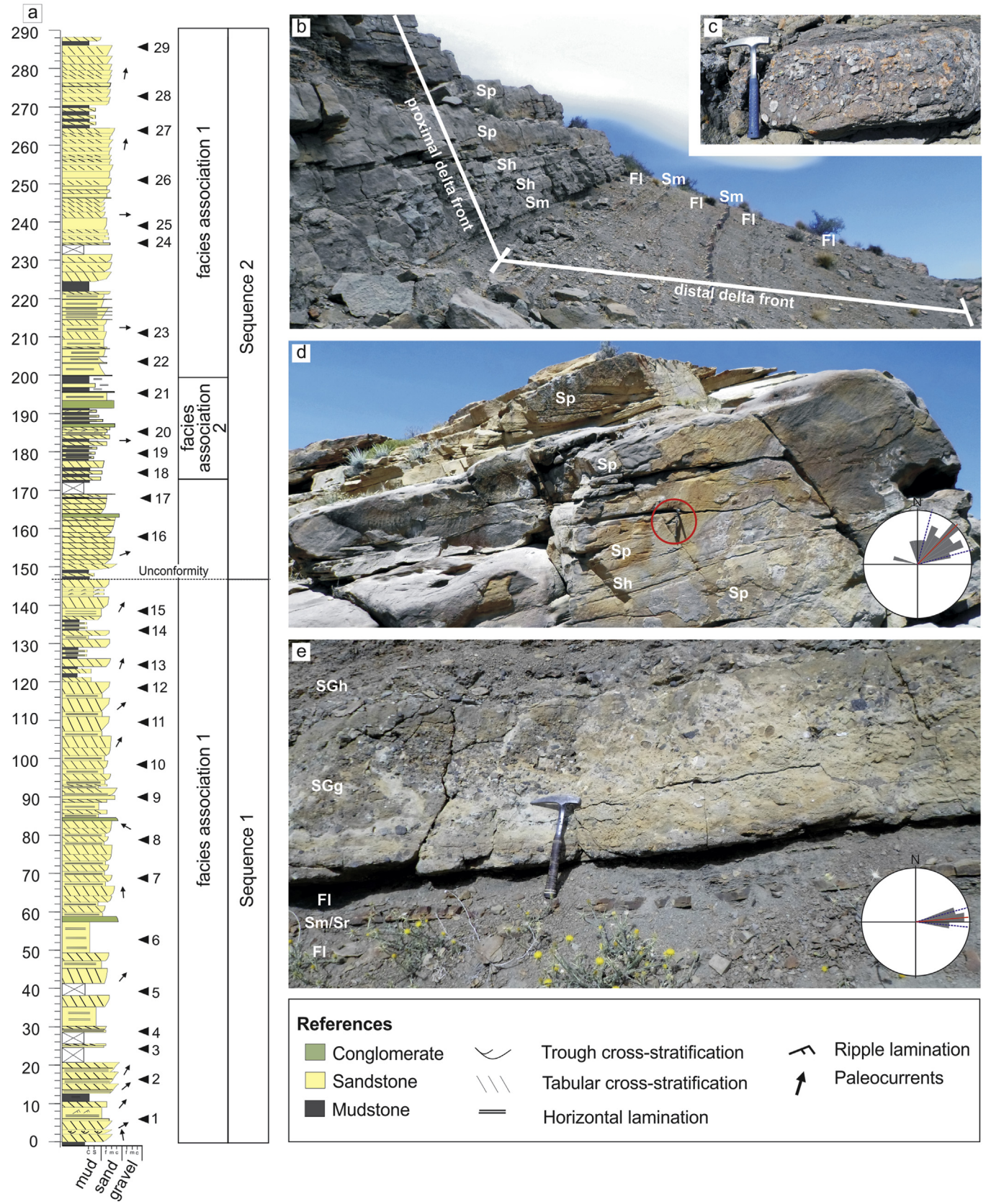


Fig. 3. a) Litholog with position of (right) paleomagnetic sites, facies and sequences. b-d) Coarsening -upwards mouth bar successions of facies association 1, showing main lithofacies and paleocurrents. e) Hyperpycnal lobe-channel within prodelta deposits of facies association 2.

Table 1
14 facies were determined based on lithological and textural features, sedimentary structures, strata geometries, and fossil content. These were grouped in two facies associations (FA 1,2) of fluvial-dominated delta system. Check marks: corresponding facies association.

Facies	Code	Thickness (cm)	Geometry/trend	Features	Traces fossils	Sedimentary process	FA1	FA2
Laminated mudstones	Fl	20 to 280	tabular	rhythmic alternation of clayey and silty laminae	*Planolites, Paleophycus, Teichicnus*	Fall-out from suspension and hyperpycnal flows (under flows)	✓	✓
Horizontal laminated sandstones	Sh	50 to 300	tabular	Microflute casts	*Paleophycus, Gyrochorte*	Turbidity currents or hyperpycnal flows (under flows)	✓	✓
Ripple laminated sandstones	Sr	5 to 10	tabular		*Paleophycus, Gyrochorte, Thalassinoides*	Ripple modification of mouth bars/turbidity currents	✓	✓
Planar cross-stratified sandstones	Sp	30 to 200	tabular	Collapse structures, convolute stratification	*Ophiomorpha, Gordia, Gyrochorte*	Distributary mouth bars	✓	
Trough cross-stratified sandstones	St	30 to 200	tabular	Collapse structures, convolute stratification	*Ophiomorpha, Gordia, Gyrochorte*	Distributary mouth bars	✓	
Massive sandstones	Sm	30 to 80	tabular	adhesion warts		Top of distributary mouth bars	✓	
Bioclastic massive sandstones	Sbm	20 to 80	tabular to lentiform	erosive bases	–	Top of distributary mouth bars	✓	
Planar cross-stratified conglomeradic sandstones	SGp	50 to150	tabular	erosive bases	–	Distributary mouth bars	✓	
Planar cross-stratified sandy conglomerates	GSp	50 to150	tabular	erosive bases	–	Distributary mouth bars	✓	
Massive to graded conglomerates	Gm/g	50 to100	tabular to lentiform	erosive bases	–	Hyperpycnal lobe channel		✓

Table 2
AMS parameters from Portada Covunco. Symbols: k_m = site mean susceptibility, Pj: anisotropy degree, L: lineation, F: foliation, K_1 (K_3) maximum (minimum) axes; Dec_t, Inc_t: mean direction after bedding correction.

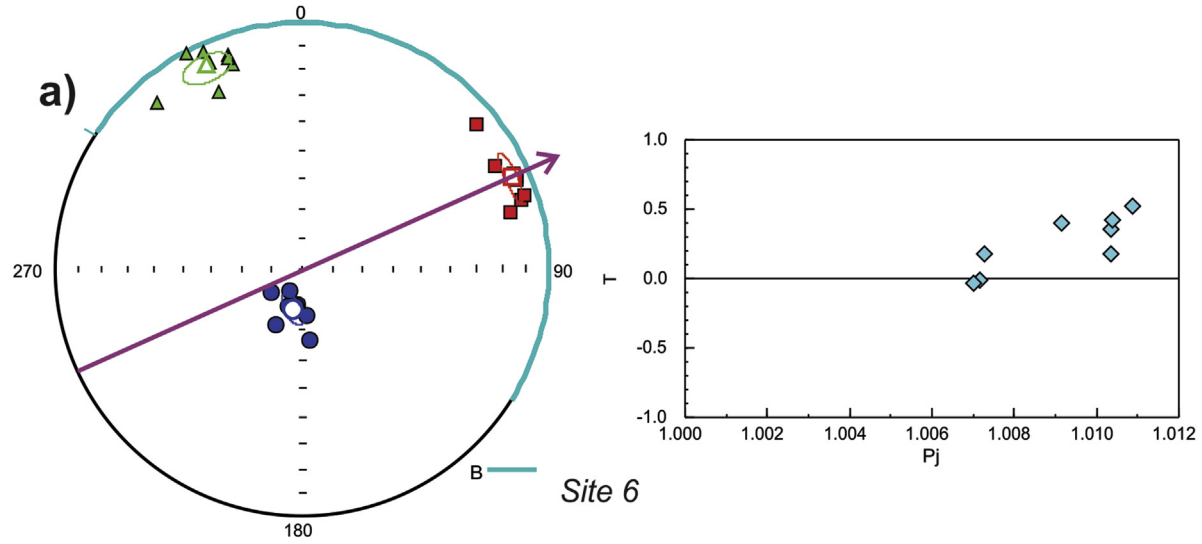
Site	k_m	Pj	F	L	K_1 Dec_t, Inc_t	K_3 Dec_t, Inc_t
1	2.66E-04	1.015	1.005	1.010	56.0, 1.8	295.8, 86.3
2	3.52E-04	1.015	1.005	1.010	261.6, 13.6	122.2, 72.4
3	2.54E-04	1.006	1.002	1.004	373.1, 28.3	160.0, 36.1
4	3.63E-04	1.015	1.006	1.009	257.0, 3.3	132.3, 84.2
5	2.61E-04	1.015	1.003	1.011	77.2, 7.6	169.3, 15.5
6	1.84E-04	1.009	1.006	1.003	66.3, 8.2	192.8, 76.4
7	3.46E-04	1.014	1.007	1.006	65.0, 2.0	165.8, 79.5
8	3.25E-04	1.009	1.005	1.004	71.5, 3.0	192.8, 84.2
9	3.89E-04	1.008	1.001	1.006	53.8, 3.1	157.2, 76.8
10	3.52E-04	1.014	1.008	1.006	258.5, 14.1	81.4, 75.9
11	2.34E-04	1.010	1.005	1.005	258.3, 3.1	61.0, 86.7
12	2.36E-04	1.010	1.003	1.007	66.3, 1.5	170.5, 84.0
13	2.61E-04	1.007	1.002	1.005	281.7, 8.2	144.6, 78.9
14	2.86E-04	1.014	1.009	1.006	75.0, 6.7	197.4, 77.6
15	2.63E-04	1.013	1.009	1.004	73.0, 6.7	180.2, 68.3
16	3.23E-04	1.009	1.001	1.007	75.2, 3.6	183.7, 78.6
17	2.23E-04	1.013	1.006	1.008	283.0, 9.2	28.1, 58.1
18	3.34E-04	1.009	1.003	1.006	80.4, 1.4	171.3, 34.5
19	2.31E-04	1.009	1.006	1.004	69.9, 4.1	209.8, 84.6
20	3.00E-04	1.012	1.006	1.006	69.8, 11.9	208.7, 74.4
21	2.69E-04	1.027	1.011	1.015	6.0, 32.9	202.1, 56.0
23	2.11E-04	1.011	1.004	1.006	260.8, 20.4	134.2, 58.0
24	1.18E-04	1.005	1.003	1.002	255.8, 17.7	6.6, 48.1
25	2.46E-04	1.008	1.003	1.005	252.8, 6.4	144.4, 70.5
26	2.98E-04	1.010	1.004	1.006	240.3, 2.1	148.2, 44.5
27	2.32E-04	1.025	1.012	1.012	39.3, 5.4	145.7, 71.4
28	2.20E-04	1.008	1.005	1.003	91.1, 41.1	349.0, 71.0
29	1.97E-04	1.005	1.002	1.003	59.3, 4.4	234.7, 85.6

Paleomagnetic treatment involved both alternating field (AF) and thermal (TH) demagnetization methods, carried out at the Laboratório de Paleomagnetismo do Departamento de Geofísica – Instituto de Astronomia, Geofísica e Ciências Atmosféricas, Universidad de Sao Paulo, Brasil. For demagnetization treatments, a TD 48 Schoensted oven and a 2G AF demagnetizer were used. Residual remanent magnetization was measured with a cryogenic 2G DC SQUID magnetometer.

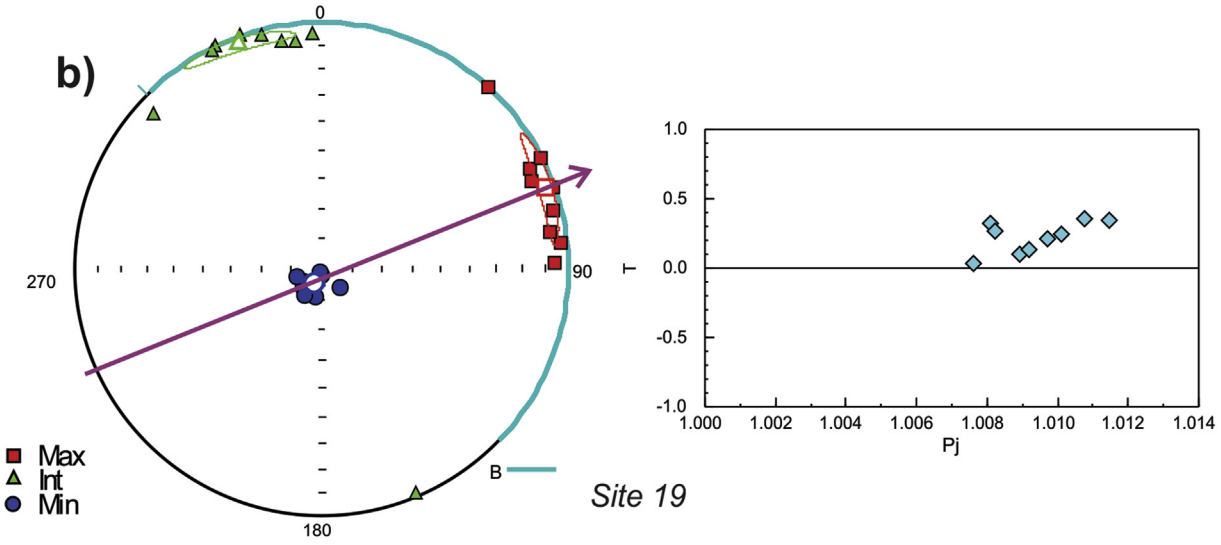
Anisotropy of magnetic susceptibility (AMS) and rock magnetic studies were performed at the Laboratorio D. Valencio, IGEBA. For AMS, measurements were performed with an AGICO MFK1A Kappabridge equipped with an automatic spinner that produces 64 measurements on each of the three axes of the specimen. For rock magnetic studies, isothermal remanent magnetization (IRM) acquisition and thermomagnetic (bulk susceptibility versus temperature) curves were obtained. For IRM acquisition curves, samples from representative lithologies were subjected to a 13 steps sequence, using an IM-10-30 pulse magnetizer (ASC Scientific) to 1T. After each step, the remanence was measured with a JR6 Dual Speed Spinner Magnetometer (AGICO inc.). Thermomagnetic curves were performed in Ar atmosphere measured in representative specimens using the AGICO MFK1-FA Multi-Function Kappabridge coupled to high-temperature (CS4) and low-temperature (CSL) controllers. The amplitude of the applied alternating field was 200 A/m, at a frequency of 997 Hz. At low temperatures the range covered was from −188 °C to 5 °C and at high temperatures from 30 °C up to 680 °C.

5. Anisotropy of magnetic susceptibility

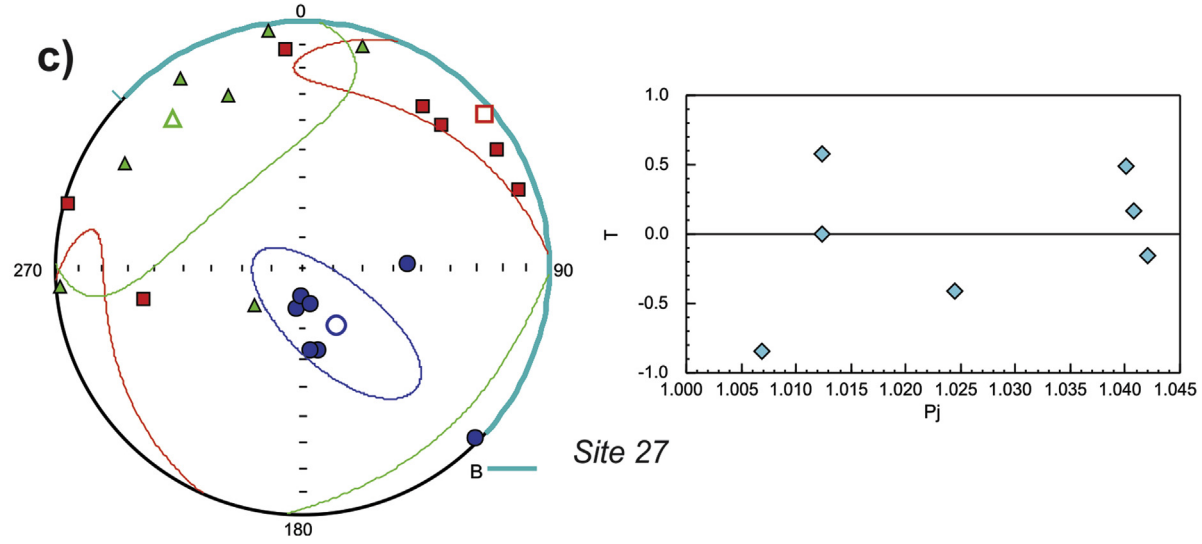
AMS studies the petrofabrics of rocks, and depends on the anisotropy of individual grains and their spatial arrangement in a rock. Magnetic properties vary with the direction, determining different types of anisotropy (Borradaile and Henry, 1997a,b; Borradaile and Jackson, 1993; Lanza and Meloni, 2006). AMS data record the preferred orientation of anisotropic magnetic minerals, i.e. paramagnetic, diamagnetic and ferromagnetic minerals that make up the rock. The



Site 6



Site 19



Site 27

(caption on next page)

Fig. 4. Lower hemisphere equal area projection of adjusted AMS axes. **a), b)** AMS fabric shapes are oblate to slightly triaxial, with girdle-distributed K_1 and K_2 on the bedding plane (green line) and K_3 slightly imbricated, suggesting primary sedimentary petrofabrics. K_1 axes clustered in the NE quadrant result from the accommodation of grains on the leeward face of cross-stratified ripples indicating that the paleocurrent flowed from SW to NE. c) In a few sites, the influence of waves produces the inconsistent rotation of grains disturbing the original sedimentary fabrics. Symbols: Maximum (square), intermediate (triangle) and minimum (circle) axis. B: bedding plane. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

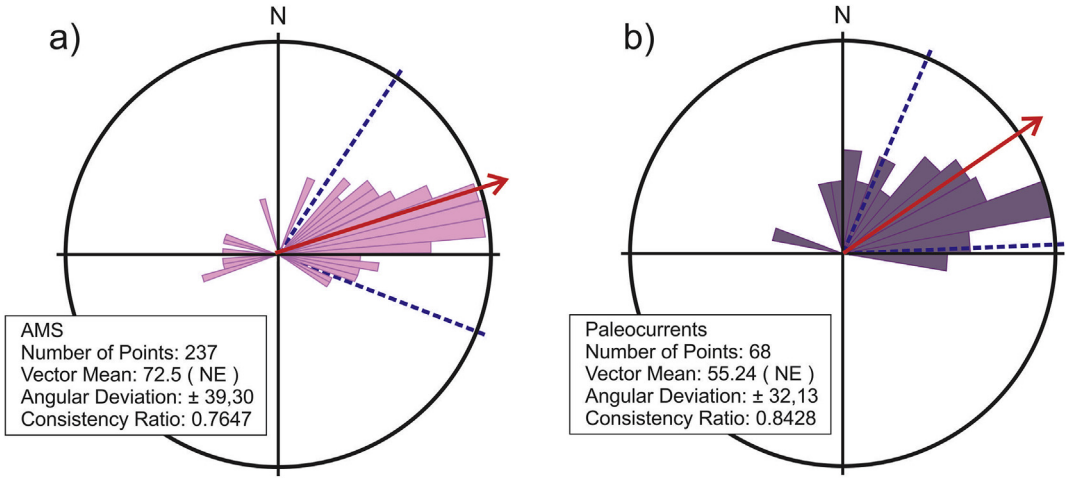


Fig. 5. Rose diagrams showing paleocurrents in the studied section. a) determined from AMS studies and b), from sedimentary structures, indicate that they flowed from SW to NE. Solid yellow line: mean direction, dashed blue lines: angular dispersion. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

susceptibility of anisotropic samples is a second-order tensor that is represented by three principal susceptibility axes (K_1 maximum, K_2 intermediate, K_3 minimum).

The anisotropy magnitude and orientation of the magnetic susceptibility ellipsoids ate Portada Covunco were calculated according to Jelinek (1978). Site mean magnetic susceptibility (km) ranges (Table 2) between 1.18×10^{-4} and 3.89×10^{-4} SI, the degree of anisotropy (Pj) is very low and ranges between 1.027 and 1.006, being highest at sites 21 and 27 where Pj reaches up to 2.7%. Foliation (F) values range from 1.012 to 1.001 and lineation (L), from 1.015 to 1.002. Observed fabric shapes are oblate to weakly triaxial with slightly girdle-distributed k_1 and k_2 axes on the bedding plane and subvertical to variably tilted K_3 axes, and maximum K_1 axes are clustered in the NE quadrant (Fig. 4, Table 2). These fabrics may be the result of suspension settling of grains from still water or a weak current (Park et al., 2013).

In sedimentary rocks such as those from Portada Covunco, it is expectable to find three petrofabrics at least (e.g. Hrouda, 1982a,b, Rochette et al., 1992; Borradaile and Henry, 1997a,b; Parés et al., 1999): primary or sedimentary, secondary or tectonic and mixed -the superposition of the primary and secondary-. In the case of non-deformed primary sedimentary fabrics, the susceptibility ellipsoid is strongly oblate, with the minimum K_3 axes orthogonal to the bedding plane, a distinct magnetic foliation and a poorly defined or absent magnetic lineation. Paleocurrents affect AMS two-fold: platy grains tend align with the flow, so that foliation plunges either upstream in the case of imbrication, or downstream when grains accommodate on the leeward slip face. Therefore, imbrication of the magnetic foliation gives the paleocurrent direction, and individual specimens lineation form a girdle and often cluster close to the flow direction. In the presence of a current, the distribution of lineation results from the compromise between bedding slope and flow velocity (Lanza and Meloni, 2006). On the other hand, the influence of waves can rotate grains in a complex fashion producing a remarkable dispersion (e.g. Liu et al., 2001).

This distribution of the K_1 axes can result from either the action of paleocurrents, or tectonic stresses (Sagnotti et al. 1994; Borradaile and Henry, 1997a,b; Parés et al., 1999; Lanza and Meloni, 2006). The sampled section is part of NW-SE fold whose axis (Fig. 2) is orthogonal to the maximum stress -σ1- direction (Yagupsky pers. comm. 2017). This means that σ1 would be in this case parallel to the K_1 axis, in opposition to the orthogonal direction that K_1 is expected to present with respect to σ1, should the fabric be tectonic (Sagnotti and Speranza, 1993; Sagnotti et al., 1994; Borradaile and Henry, 1997a,b). Therefore, the alignment of the K_1 axes in this case as well the imbrication of the K_3 axes are more likely due to the action of paleocurrents. In the studied section, the clustering of K_1 axes, most likely displays the accommodation of grains on the leeward slip face of current ripples that aligns the fabrics with the flow. The vector mean of paleocurrents determined from AMS analysis is 73° (NE) with angular deviation of ± 39° (Fig. 5a). Therefore, according to AMS data paleocurrents would have flowed from SW to NE, which agrees with that measured in the area (Fig. 5b), from cross stratified facies (e.g. Zavala, 1993, 1996a,b).

6. Rock magnetic studies

Thermomagnetic curves (bulk susceptibility k versus temperature T) and isothermal remanent magnetization (IRM) acquisition curves were carried out in representative lithologies of the Lajas Formation at Portada Covunco. The goal was to provide a better characterization of the remanence carriers.

6.1. Thermomagnetic curves

Thermomagnetic curves are an effective tool to identify the minerals that carry the magnetic remanence (Hrouda, 1994), thus high and low temperature curves were performed in representative lithologies. One of the most conspicuous feature is the irreversibility of the curves that indicates the neoformation of high susceptibility minerals –probable titanomagnetites-from paramagnetic minerals (Fig. 6a). In addition, low temperature and heating curves evince (Fig. 6b and c) a high concentration of paramagnetic minerals, particularly observed from the 1/T hyperbole of low temperature curves (Fig. 6a,b,c). On the other hand, heating curves denote an increase in susceptibility between 530 and 540 °C defining Hopkinson peaks indicative of the presence of SD Ti-poor magnetite. Although we are not sure whether this magnetite corresponds to the original of the sediments, or a new mineral formed during the heating process, we interpret that is most likely the latter

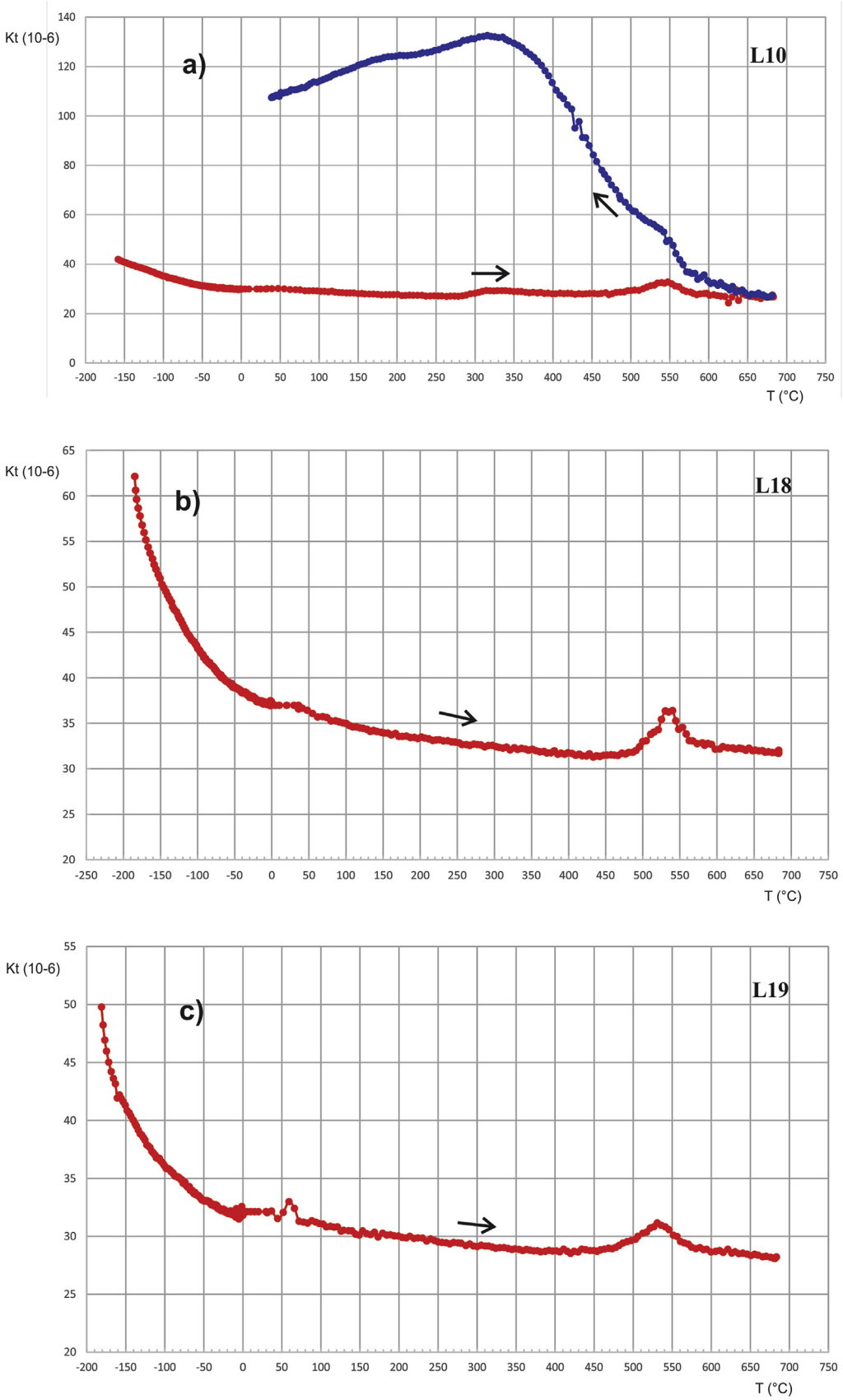


Fig. 6. Thermomagnetic curves in representative lithologies. a) All of them are irreversible, indicating the neoformation of high susceptibility magnetic minerals –titanomagnetites-. b), c) the low temperature curves remark the abundance of paramagnetic minerals, whereas the heating curves show the presence of Hopkinson peak at 530–540 °C suggesting the occurrence of Ti-poor magnetite.

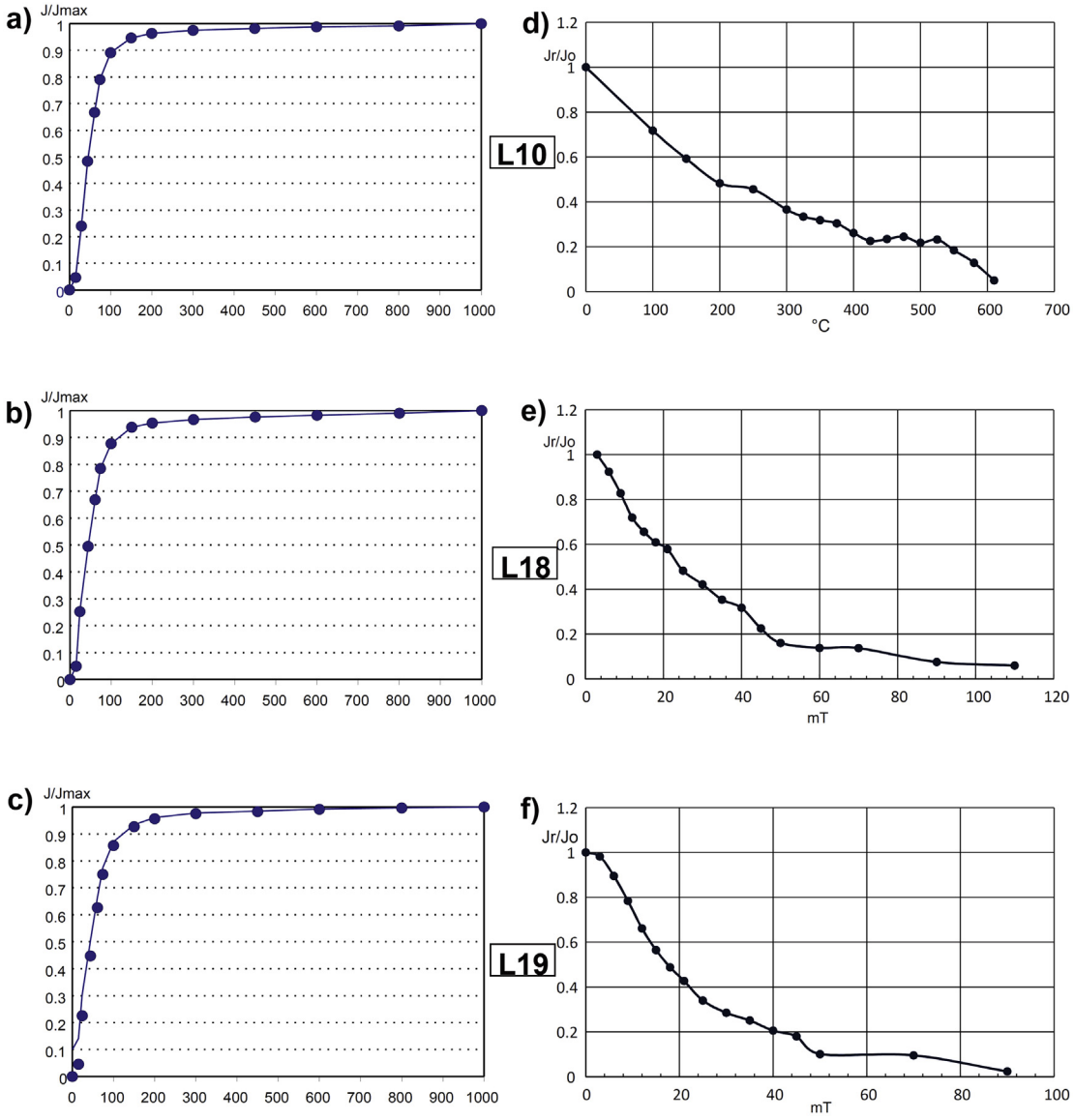


Fig. 7. a), b), c) IRM acquisition curves of sister specimens, showing the presence of a high (hematite?) and a lower coercivity mineral (magnetite), and d) thermal, e), f) AF demagnetization diagrams suggest that the remanence is carried primarily by Ti-poor magnetite.

case due to the typical low susceptibility values of samples.

6.2. Isothermal remanent magnetization (IRM)

IRM curves acquisition in sister specimens (Fig. 7a,b,c) show that specimens have not reached saturation at 1T. Rather, IRM indicates the presence of high and lower coercivity minerals. They characterize most likely, the coexistence of magnetite plus an antiferromagnetic mineral (hematite?).

On the basis of thermomagnetic curves, IRM acquisition curves and demagnetization diagrams (Fig. 7d,e,f) we interpret that the remanence is carried mostly by Ti-poor magnetite, and secondly by hematite? Moreover, AF demagnetization diagrams (Fig. 7e and f) would indicate that the magnetite is typical PSD according to Dunlop et al. (1997).

7. Paleomagnetic analysis

Demagnetization treatment involved at least 16 steps for TH and AF methods (from 100 to 640 °C and from 3 to 110 mT, respectively). After each thermal demagnetization step, magnetic susceptibility (k was monitored to prevent formation of new magnetic minerals. Initial k ranged between 1.18×10^{-4} to 3.89×10^{-4} SI.

Paleomagnetic analysis was performed using software designed at the Utrecht University, The Netherlands. The software used to calculate magnetic components and mean site directions, as well as paleomagnetic poles is also available at Paleomagnetism.org website (Koymans et al., 2016).

Two magnetic components were systematically removed along the section. The softer component was erased at either c.250 °C or < 15 mT, bearing mostly northern declinations with negative inclinations (Figs. 8 and 9). The harder component was isolated between 250 °C and 550 °C or 15–50 mT. This component is interpreted as the characteristic remanent magnetization (ChRM) of the section and when corrected to the paleohorizontal, directions show NE (SW) declinations with negative (positive) inclinations (Figs. 8 and 9).

Isolated ChRM showed two principal behaviours: a) straight trajectories to the origin (Figs. 8a and 9a), and b) curved trajectories defining great circles (Figs. 8b and 9b). Finally, less than 5% of the specimens revealed unstable or unreliable behaviours. Components of specimens type a) were calculated using principal component analysis (PCA, Kirschvink, 1980) with the Paldir software. This kind of behaviour, which has been observed in a few specimens from mainly normal

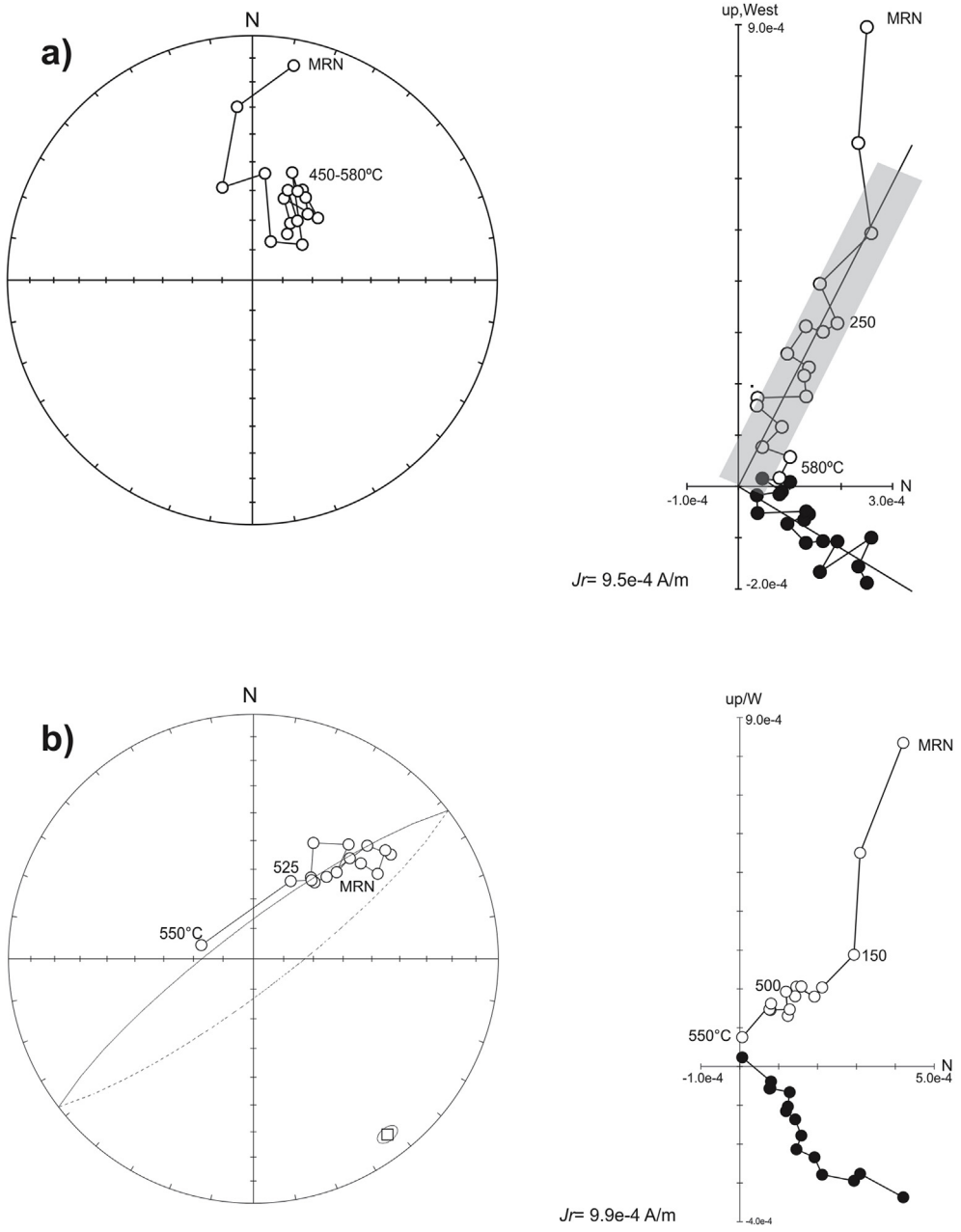


Fig. 8. Representative orthogonal plots and stereoplots (bedding-corrected) from the Lajas Formation. The examples corresponds to site 3) bearing normal polarity. **a, b)** shows two components with negative inclinations, one soft removed at < 250 °C/15 mT, and the ChRM isolated between 250 °C/90 mT, calculated with PCA (Kirschvink, 1980). Such behaviour was combined with remagnetization circles to obtain mean site directions. Symbols: full dot = declination; open dot = apparent inclination. *Jr* = NRM intensity in Ampere/meter.

polarity sites (Table 3), were considered as reference directions for the calculations of remagnetization circles. Thus, mean site directions from components calculated with PCA (i.e normal polarity sites, Table 3) were derived using fisherian statistics (Fisher, 1953) with the Palfish software Mean directions from sites with remagnetization circles, were in all cases, calculated using the combination of remagnetization circles with reference directions calculated from PCA (i.e reverse polarity directions, Table 3). In the latter case, the Palfit software was used based on (Halls (1976) and McFadden and McElhinny (1988). For this study, sites with N ≥ 4, $^{\alpha}95 = \leq 15^{\circ}$ and k ≥10 were included in the analysis, thus sampling site 22 was ruled out (Table 3).

The mean direction of the soft component *in situ* is Dec. = 9.3°, Inc. = −41.5°, α_{95} = 2.1°, K = 19.8, N = 241 and Dec = 350.5°, Inc. = −59.4°, α_{95} = 2,5°, K = 14.2, N = 241, when corrected to the paleohorizontal (Fig. 10). The mean direction (yellow circle, Fig. 10) with tectonic correction is almost coincident with the Late Cretaceous reference direction (red triangle, Fig. 10a) for South America (Somoza and Zaffarana, 2008). Thus, it could represent a chemical remagnetization acquired during the first stages of formation of the fold and thrust belt due to fluids migration (e.g. McCabe and Elmore, 1989; Zegers et al. 2003). On the other hand, when presented *in situ* coordinates, the mean direction is fully consistent with the present-day field at Dec = 4.6°, Inc = −39.3° (green star, Fig. 10b), which alternatively, could represent a viscous imprint on MD or large PSD grains.

The mean direction of ChRM, when considered in reverse polarity, bears an *in situ* direction at Dec = 208.3°, Inc = 8.5°, α_{95} = 4.3°, K = 41.4, N = 28, and Dec = 207.2°, Inc = 34.0°, α_{95} = 3.8°, K = 52.5, N = 28 when tectonic corrected (Fig. 11c). The statistical

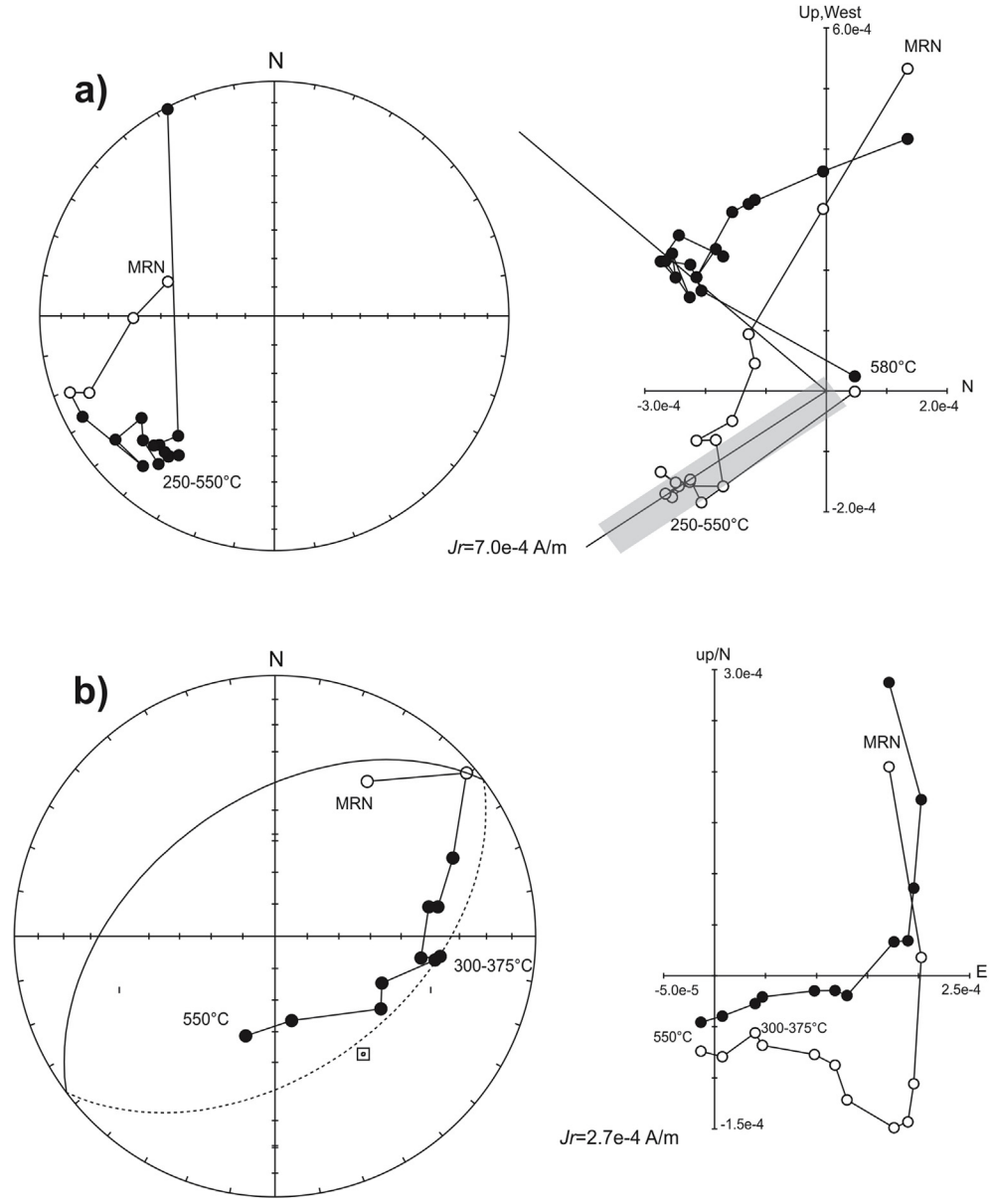


Fig. 9. Representative orthogonal plots and stereoplots (bedding-corrected) from sites with reverse polarity. a) site 29 shows two components, one soft with negative inclination removed at < 250 °C/15 mT, and the ChRM with positive inclinations isolated between 250 °C/90 mT, calculated with PCA (Kirschvink, 1980). This was combined with remagnetization circles to obtain mean site directions. **b)** remagnetization circle site 18. Symbols as in Fig. 8.

parameters of the *in situ* is almost horizontal mean direction, clearly improve when corrected to the paleohorizontal (Fig. 11a and b) and that is a strong support of the pretectonic origin of the ChRM.

A primary approach is graphic showing that the mean normal and reverse polarity directions are statistically undistinguishable when corrected to the paleohorizontal (Fig. 11c). When performing the statistical test from McFadden and McElhinny (1990), statistical parameters are: N = 28, critical angle = 15.1°, angle between polarities = 13.7°, conf. at 95% = 9.1 and at 99% = 15.7, test value = 7.6, which classifies it as R_C. The positive test indicates that normal and reverse directions are antiparallel and thus, that the systematic secondary component has been successfully removed. In addition, since normal and reverse polarity directions belong to the same stratigraphic section, we interpret that the magnetization is primary.

8. Paleomagnetic pole

A paleomagnetic pole (PP) for the Middle Jurassic –Bathonian- Lajas Formation calculated from virtual geomagnetic poles (VGPs), is located at 167.1 °E, $^{59.4}_{95}$ °S, N= 28, A= 3.6°, K= 57.2 (Table 4). This PP was compared (Table 4) with others available for the Middle to Upper Jurassic of stable South America (Fig. 12) to test whether the studied section was subjected to rotations around vertical axis and/or inclination shallowing. Among this group of paleopoles, only one is 170 Ma (K/Ar), i.e. Bathonian, obtained at Mamil Choique (North Patagonian Massif) from basic to intermediate dykes (Rapalini and López de Luchi, 2000). The ChRM of this pole passes both a baked contact and reversal test, so that the magnetization is interpreted as primary. However, since Mamil Choique is located close to the Gastre Fault System (Rapela and Pankhurst, 1992), the paleopole may be slightly rotated. The Portada Covunco PP falls (Fig. 12) close to that of Mamil Choique although at a lower latitude, which is strongly indicative of inclination shallowing of the magnetic vector because of mechanical compaction (King, 1955; Anson and Kodama, 1987; Deamer and Kodama, 1990; Kodama, 1997; Tan and Kodama, 2002; Tan et al. 2010). Using the PP of Mamil Choique as reference, we calculated: R = 4° ± 10°, p = 14° ± 8°, and

Table 3
Paleomagnetic data of sampling sites. Symbols: n/N = total/selected samples; ChRM = characteristic remanent magnetization; D_t, I_t = declination, inclination of ChRM after bedding correction; α_{95} = semi-angle of confidence; *K* = Fisher's precision parameter; VGP = virtual geomagnetic pole; Bedding correction = Strike follows the right-hand rule. In italics, site mean directions with sufficient PCA components (Kirschvink, 1980) were calculated using fisherian statistics (Fisher, 1953). Other mean site directions were obtained from the combination of remagnetization circles and reference PCA components (Halls, 1976; McFadden and McElhinny, 1988).

Site	n/N	*X* (10^{-4} SI)	ChRM		$^{o}95$ (°)	*K*	VGP		BP
			^{Dec}t	^{Inc}t			Lat. °S	Long. °E	
1	11/11	2.66	206.0	31.7	8.5	24.7	−58.6	163.3	285/24
2	10/10	3.52	208.6	36.0	12.1	13.4	−59.0	170.5	298/31
3	*14/8*	*2.54*	*25.5*	*−56.9*	*11.2*	*19.5*	*70.0*	*203.8*	*308/24*
4	10/8	3.63	205.7	39.2	10.7	21.4	−62.5	170.0	287/30
5	9/6	2.61	211.8	24.8	10.0	32.7	−51.8	165.8	286/27
6	*8/7*	*1.84*	*194.0*	*40.9*	*10.4*	*25.9*	*70.5*	*151.4*	*298/19*
7	7/5	3.46	224.2	31.8	6.6	90.0	−46.0	183.0	287/15
8	5/5	3.25	197.6	27.0	6.6	90.0	−61.0	146.8	282/29
9	10/10	3.89	200.3	37.0	14.6	9.2	−64.8	159.3	308/23
10	8/8	3.52	209.7	30.0	11.7	17.9	−55.5	166.8	310/36
11	10/9	2.34	204.7	35.0	13.9	11.3	−61.1	164.3	308/31
12	9/7	2.36	200.3	26.9	11.1	22.7	−59.6	151.3	315/31
13	8/5	2.61	203.3	26.5	8.5	54.3	−57.8	155.7	305/31
14	11/6	2.86	209.6	25.6	7.9	52.3	−53.6	163.7	320/21
15	12/10	2.63	198.6	29.1	4.8	85.1	−61.6	149.9	311/20
16	11/10	3.23	204.5	33.1	10.7	17.1	−60.3	162.3	315/22
17	10/5	2.23	216.0	28.9	12.5	25.1	−50.7	173.2	302/34
18	10/5	3.34	221.6	27.2	13.4	21.8	−46.1	177.6	305/24
19	*11/7*	*2.31*	*25.0*	*−52.4*	*12.5*	*17.9*	*69.0*	*191.2*	*305/24*
20	*15/13*	*3.00*	*40.1*	*−27.8*	*11.1*	*12.2*	*47.4*	*176.6*	*310/21*
21	6/5	2.69	213.0	35.3	15.0	17.4	−55.6	175.0	308/22
22	4/3	1.64	148.3	28.6	25.6	10.0	–	–	304/27
23	7/6	2.11	206.4	32.1	11.3	25.6	−58.6	164.1	290/19
24	7/6	1.18	197.8	37.8	13.3	18.5	−66.7	155.7	307/34
25	16/12	2.46	198.9	42.3	11.6	12.1	−68.5	163.3	307/34
26	*7/4*	*2.98*	*5.9*	*−42.1*	*12.7*	*30.4*	*74.59*	*130.4*	*309/26*
27	9/7	2.32	198.6	26.2	9.7	29.8	−60.1	148.0	308/27
28	9/7	2.20	225.3	28.9	5.0	112.0	−44.0	182.0	308/27
29	12/5	1.97	220.6	29.6	8.0	61.3	−47.8	178.3	298/20
PP	**N**		**$^{A}95^{(°)}$**	**Lat(°S)**	**Lon(°E)**				
	28	-	**3.6**	**59.4**	**167.1**	-	-	-	-

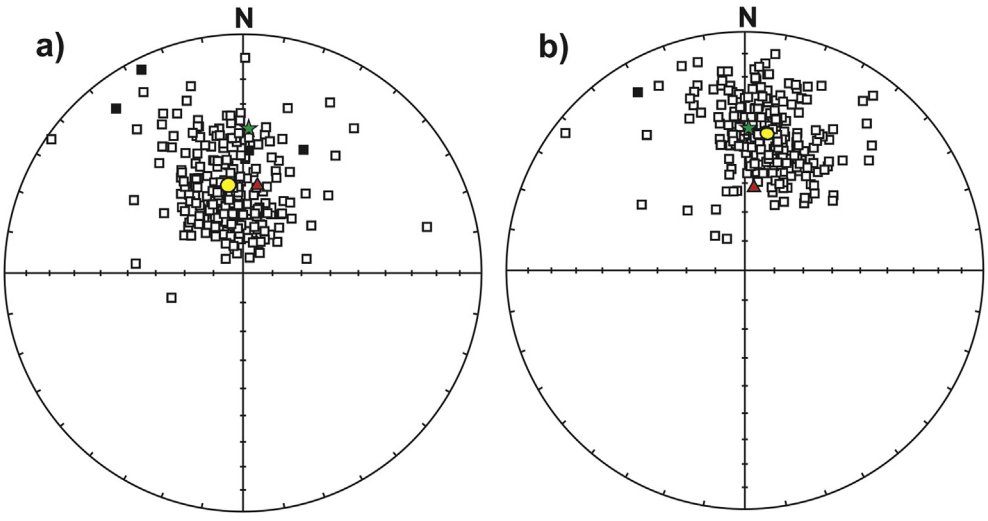


Fig. 10. a) tectonic corrected direction from the soft component bearing northern declinations and negative inclinations yield (yellow circle) coincides with the that of the Late Cretaceous (Somoza and Zaffarana, 2008) (red triangle), and **b)** the *in situ* mean direction falls very close to the present-day magnetic field. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

F = 18° ± 11°, which implies that the section at Portada Covunco has not been significantly rotated but there is flattening of the magnetic vector. Shallowing inclination caused by compaction can be expressed as tan (I_r) = *f* tan (I_o), where I_r is the reference (field) inclination, I_o is the observed (measured) inclination and *f* is the shallowing factor (King, 1955). In this study, I_r = 52.6° and I_o = 34° so that *f* = 0.52, in agreement with those expected by King (1955), Tauxe and Kent (1984) and the minimum *f* for magnetite-bearing rocks = 0.54 with standard deviation = 0.08 from Bilardello and Kodama (2010). On the other hand, considering that the amount of inclination shallowing is the difference between I_r and I_o = 19°, compaction (ΔV) of the sequence is = 30% (Arason and Levi 1990), in agreement with the amount of compaction expected for medium to fine-grained sandstones (e.g. Beard and Weyl, 1973; Scherer, 1987).

Using the PP from Mamil Choique, the paleolatitudinal position paleolatitude of Portada Covunco during the Bathonian was 33°S. Such paleolatitude is consistent with palynologic studies from the Lajas Formation at Portada Covunco indicate that during the Late Bajocian-

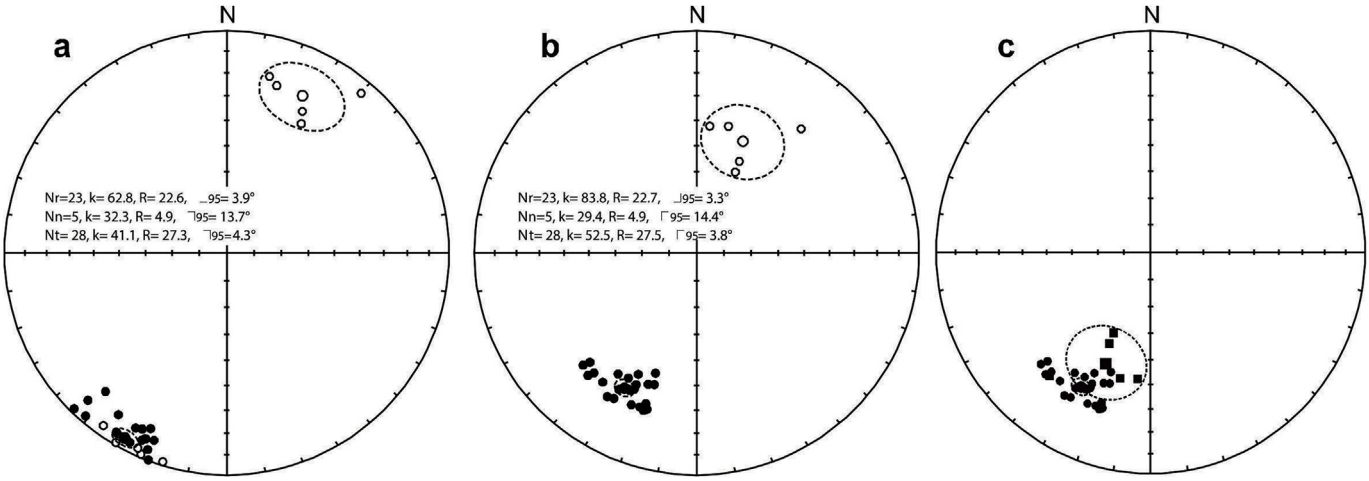


Fig. 11. Mean site directions a) *in situ* and b) corrected to the paleohorizontal. Statistical parameters improve in the latter case which suggest that the isolated ChRM was acquired prior to the tectonic deformation. c) Reverse (circles) and normal (squares) polarity site directions are stastically undistinguishable, which indicates that they pass the reversal test. Symbols as in Fig. 8.

Table 4
Middle Jurassic paleomagnetic poles for South America (Fig. 12).

Pole	Number	Age (Ma)	Lat (°S)	Lon (°E)	A_{95}	Reference
El Quemado Complex	1	155	81.0	172.0	5,5°	Iglesia Llanos et al. (2003)
Puerta Curaco	2	145	80.0	67.0	3,5°	Amigo (2016)
Loncoche	3	154	76.0	195,5	3,5°	Iglesia Llanos et al. (2017)
Mamil Choique	4	170	70,2	190,4	9,7	Rapalini and López de Luchi (2000)
Arroyo Covunco	5	168	59,4	167,1	3,6°	*This study*

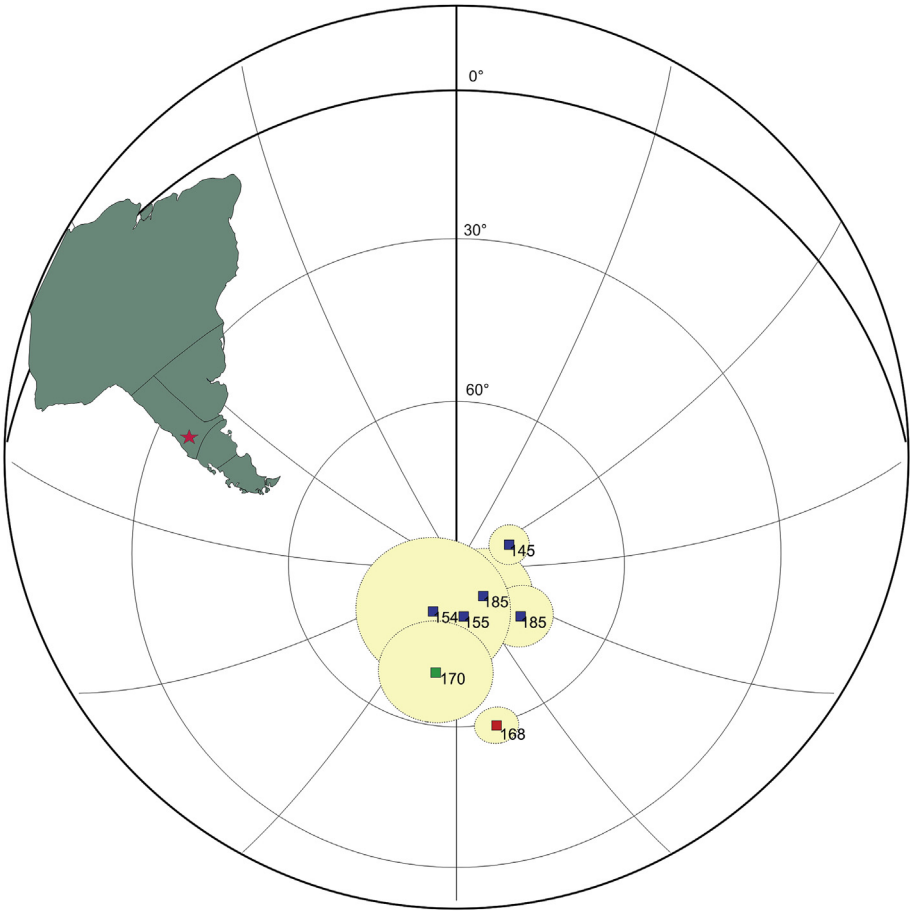


Fig. 12. Middle Jurassic paleomagnetic poles from stable South America. Red square is the PP from this study, and green square is the reference pole from Mamil Choique (Rapalini and Lopez de Luchi, 2000). With respect to the Mamil Choique, Portada Covunco falls at a similar longitude although at a lower latitude, suggesting that the studied locality was not affected by rotations around vertical axis but experienced flattening most likely produced by the c. 30°% of compaction in sandstones. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

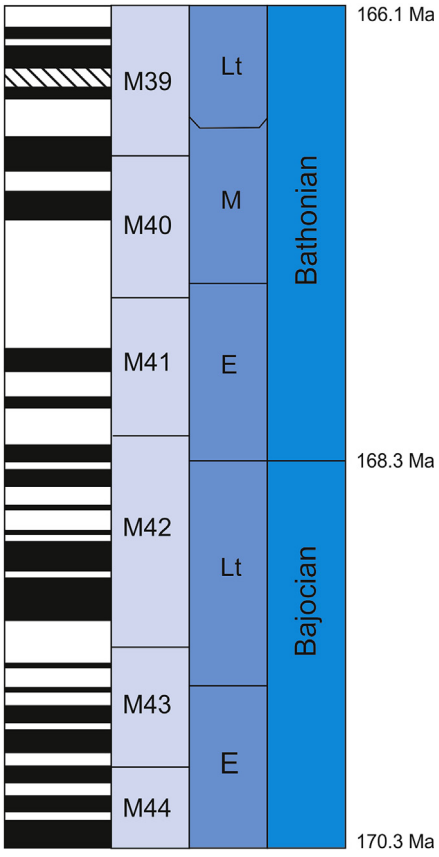


Fig. 13. Geomagnetic polarity time scale for the Bajocian through Bathonian obtained from on-land based sections comprises Chrons M44 to M39 Chrons (Ogg et al., 2016) -

Early Bathonian, warm and sub-humid climate conditions prevailed, characterized by the greatest advance of Podocarpaceae, Pteridosperm and Araucariaceae warm forests (Martínez et al., 2002).

9. Magnetostratigraphy

In the Geomagnetic Polarity Time Scale, the Bajocian through Bathonian features (Fig. 13) a time interval in which the Earth's magnetic field changed polarity at a very fast rate (Chrons M44 to M39), to such a degree that Chron M39 reveals a gap with no defined ("mixed") polarities (Ogg and Hinov 2012; Ogg et al., 2016). Marine magnetic anomalies older than Chron M26 were derived from deep-tow surveys in the Pacific, display a possible set of between 50 and 100 polarity chrons arbitrarily grouped within M26 through M44 (in Ogg and Hinov 2012 and references therein). For this time lapse, the latest Geomagnetic Polarity Time Scale (GTS16) contemplates two magnetostratigraphic scales, one obtained from deep-tow magnetic data on the relative fast-spreading Japanese lineations, and the other from sections on-land calibrated with ammonites and nannofossils in the Sub-Mediterranean and Sub-Boreal realms. The first sea-floor scale is made up of dominantly reverse with many short normal polarity intervals distributed in a relatively regular fashion, whereas that on-land bears (Fig. 13) a distinct pattern of dominantly normal and dominantly reverse polarities. For instance, the base of the Lower Bajocian (upper M44-lower M43) comprises reverse, dominant normal and dominant reverse polarities in the "on-land" scale, whereas in that from the sea-floor almost reverse polarity with a few thin normal intervals is observed. Therefore, the choice of which reference polarity scale should be used to compare with Portada Covunco compelled extra caution. In this regard, Ogg and Hinov (2012) emphasize that, unlike the standard sea-surface polarity model of M0r through M25r, the interpretation of

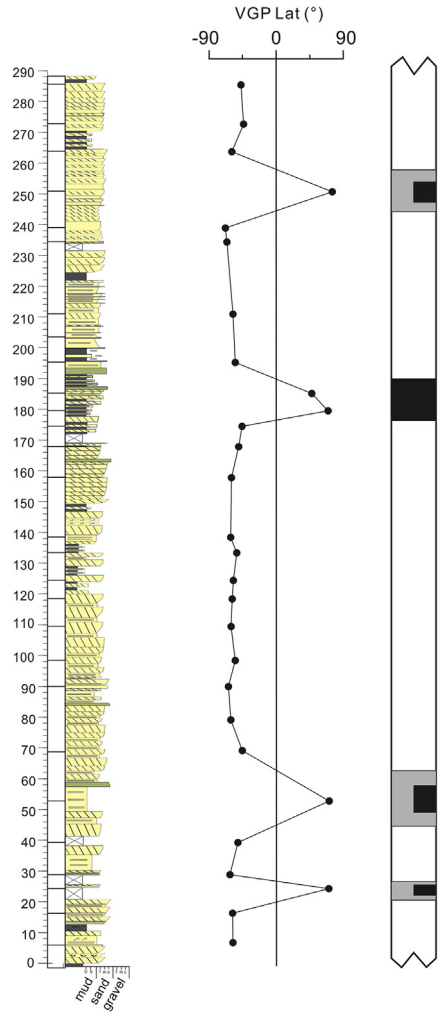


Fig. 14. Litholog, paleomagnetic sites, VGP latitudes and magnetostratigraphy from the Portada Covunco section. The magnetostratigraphy shows a dominant reverse polarity (white) with four thin normal polarity (black) zones.

the deep-tow surveys is more ambiguous. This is because signals from deep-tow surveys vary depending on considered depths, being narrow short-duration, low-amplitude polarity intervals near the oceanic crust, and with wider-width features when projected to mid-depths (Ogg and Hinov 2012). Furthermore according to these authors, on-land based magnetostratigraphies for the Oxfordian through Bajocian are indeed compatible with an intermediate geophysical model. On this basis, we regard the on-land based scale obtained for this time interval in Spainto be used as reference for Portada Covunco (Fig. 13).

At Portada Covunco, VGP latitudes define (Fig. 14) four normal and five reverse polarity zones isolated in 29 sampling sites. Polarity zones recorded from single sites are shown in as half-column within a grey zone to remark boundaries uncertainties. Thus, the base of the section bears reverse polarity (sites 1 and 2), followed by normal polarity (site 3), reverse (sites 4 and 5), normal (site 6), reverse (sites 7 to 18, *c.* 120 m.) and normal (sites 19 and 20) polarities. Upwards, follows reverse polarity (sites 21 to 25), normal (site 26) and reverse polarity (sites 27 to 29).

10. Discussion

The best fit between both magnetostratigraphic scales is achieved in the Lower through Middle Bathonian interval, bearing dominant reverse with two thin normal polarity zones at the base and other two at the top (Fig. 15). Would the local scale be moved with respect to the

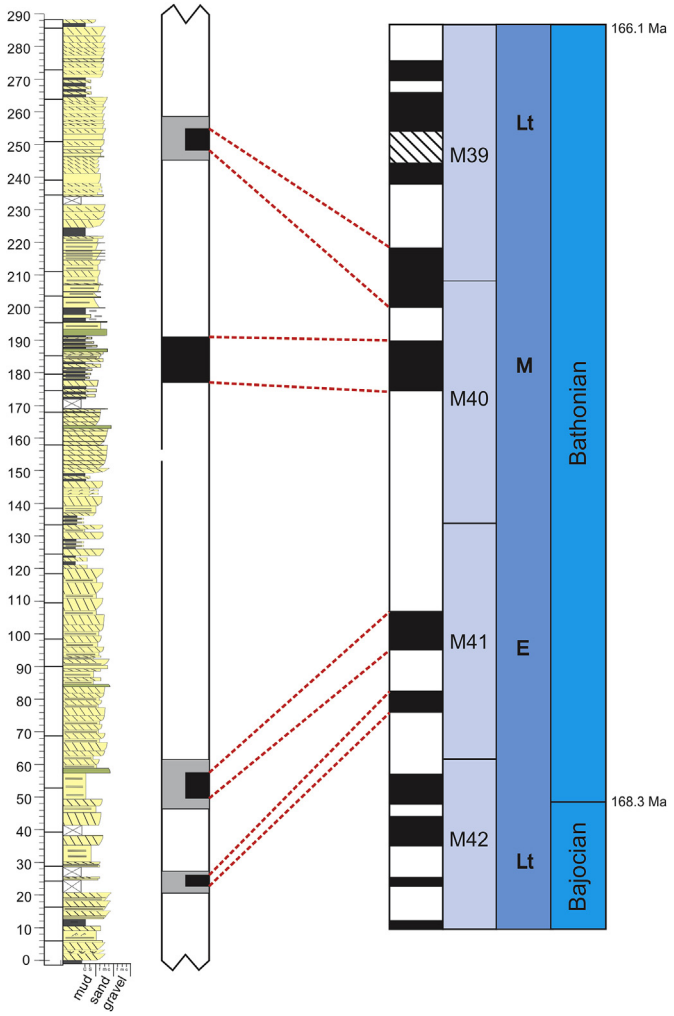


Fig. 15. Correlation between Portada Covunco and the GTS2016. We interpret that the deposition of the section started (site 1) during Chron M41 through (site 29) Chron M39, thus restricting its age to uppermost Late Bajocian to the Late Bathonian.

GTS2016 downward to the Upper Bajocian, or upward to the Upper Bathonian, the dominant polarity changes to normal (Chrons M42 and M39, respectively (Fig. 13).

Therefore, the first site within the Lajas Formation would be assigned to the base of M41-top of M42 (Lower Bathonian). Normal polarities at sites 3 and 6 would then correlate with the two normal magnetozones of M41, and the following large reverse interval to upper M41 - lower M40 that comprise the Early-Middle Bathonian boundary. The thicker normal zone at sites 19–20 is correlated with that of lower M40 and thus, the uppermost normal polarity zone is assigned to upper M40-lower M39. Upwards, the reverse polarity at the top of the section (sites 27 to 29) correspond to lower M39 of later most Middle Bathonian age (Fig. 15).

With interpreted ages of sampling sites and their stratigraphic position, we calculated (Fig. 16) a minimum sediment accumulation rate, of 170 m/Myr (Fig. 16). This rather low sedimentation rate reflects limited accommodation space, and sediment bypass, which is consistent with a progressive northward dislocation of depositional sequences described by Zavala and González (2001) for this sector of the basin. The Huincul high was an active pop-up structure during the time of deposition of the Lajas Formation, showing evidences of tectonic pulses during the Bathonian-Early Callovian that triggered the inversion of extensional basement structures (Zavala and González, 2001; Mosquera et al., 2011). Our paleomagnetic data fully support the interpretations made by these author for the Portada Covunco area and confirm that the sedimentary section does not reach the Callovian in these area as suggested by some palynological data.

11. Conclusions

The studied section of the Lajas Formation cropping out at Portada Covunco is a 290 m-thick succession made up of conglomerates, sandstones and mudstones that reaches. Two facies associations were defined within a fluvial-dominated delta system: facies association 1 interpreted as progradation of mouth bars in a delta front setting, and facies association 2 as hyperpycnal prodelta lobes.

The isolated characteristic remanent magnetization passed the reversal test as Rc (McFadden and McElhinny, 1990), indicating that no systematic secondary components biased the remanence directions. Furthermore, since normal and reverse directions belong to the same stratigraphic section, they most likely represent the primary magnetization. Based on rock magnetic and paleomagnetic studies, this component would be carried primarily, by PSD titanium-poor magnetite and possibly hematite. A systematic secondary magnetic component is interpreted as representing either remagnetization produced by the present-day magnetic field on MD or large PSD grains or, a chemical remagnetization acquired during the first stages of formation of the Agrio fold and thrust belt due to fluids migration.

Anisotropy of magnetic susceptibility studies show mostly primary sedimentary fabrics K_{max} axes group near the horizontal with a mean direction of 73° (NE) and angular deviation of 39°. Such lineation of K_{max} axes does not relate with the maximum stress σ_1 direction revealed by the regional NW-SE fold axis. Instead the clustering of the K_{max} would reflect the accommodation of the grains with on the leeward slip face of current ripples aligning the fabrics with the flow which in this case are directed to the NE (Felletti et al., 2016), in agreement with paleocurrents measured in the region from sedimentary structures.

A PP pole was calculated for Portada Covunco located at 167.1°E, 59.4°S, N = 28, A95 = 3.6°, K = 57.2. The pole was compared with the only one available for the –Bathonian (170 Ma) in South America, obtained from dykes in northern Patagonia which was used as reference. The position of the PP from Portada Covunco falls in a similar meridian than the reference poles of Mamil Choique although at a lower latitude. Using the reference PP of Mamil Choique, we calculated rotation around vertical axis (R = 4° ± 10°), paleolatitudinal placement (p = 14° ± 8°), and flattening of the magnetic vector (F = 18° ± 11°). The calculated inclination shallowing of the section yields an *f* factor = 0.53 (King, 1955) that is consistent with typical flattening values of sediments obtained from experimental data, and is coherent with the typical compaction values of sandstones that are in the order of 35%. Using the Mamil Choique PP, the locality of Portada Covunco would have been located at c. 33°S during the Late Bajocian- Early Bathonian, which is consistent with paleoclimatic interpretations derived from palynological data.

The resultant magnetostratigraphic scale bears dominant reverse with four thin normal polarity zones, denoting boundaries uncertainties. The best fit with the GTS2016 is achieved within the Lower-Middle Bathonian. Therefore, the reverse polarity at the base of the section –sites 1 to 2-would be assigned to the lower M41-upper M42 Chrons (Lower Bathonian). Up-section, normal polarity from sites 3 and 6 is assigned to the lower half of M41, the long reverse zone –sites 7 to 18- to upper M41-lower M40, the thicker normal polarity zone to the first normal polarity of M40, the last normal polarity zone to uppermost M40 - lowermost M39. At the top of the section, the reverse polarity is assigned to lower M39 (uppermost Middle Bathonian). It is important to notice that the Lajas Formation spans the whole Middle Jurassic and that the unit bears neither age diagnostic fossils, nor tuffs for obtaining absolute ages. Therefore, the magnetostratigraphic correlation applied to the unit helped to constrain its age to the Early Bathonian – early Late Bathonian in Portada Covunco area. Our paleomagnetic data

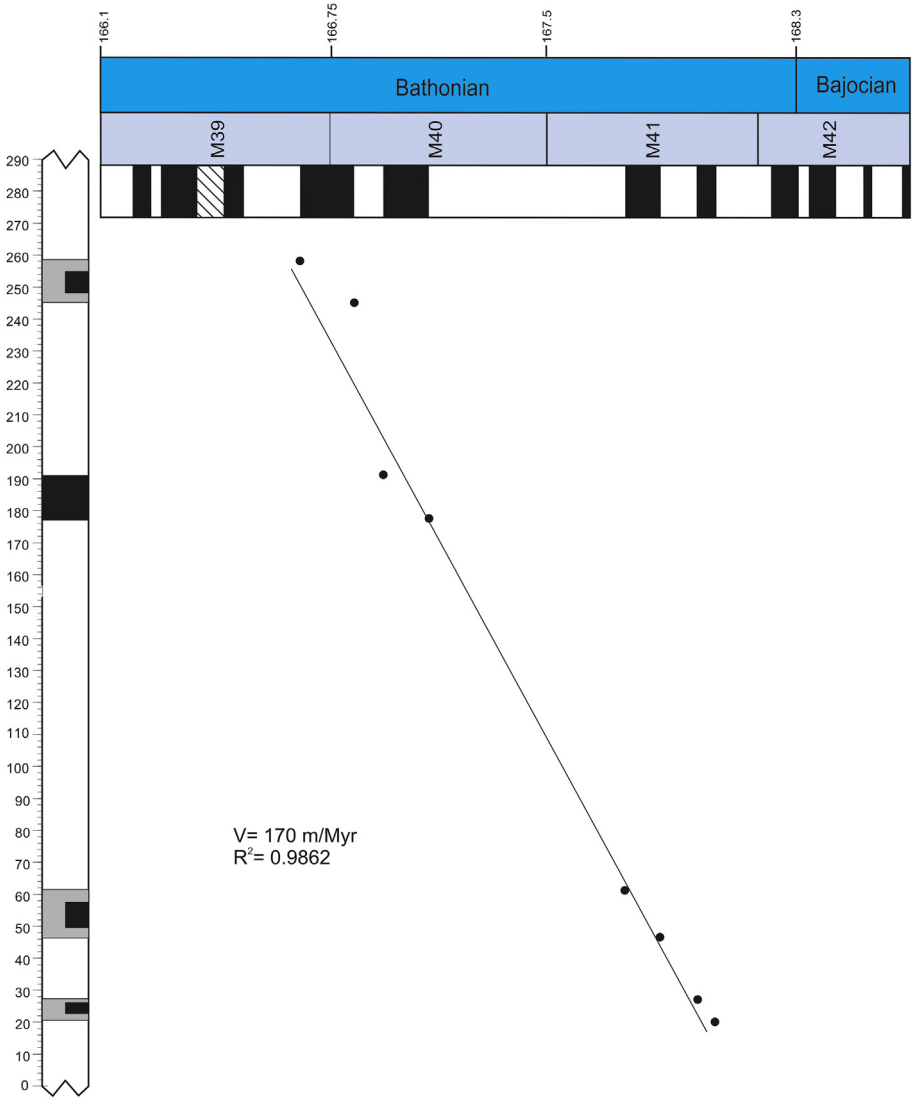


Fig. 16. A minimum sediment accumulation rate of 170 m/Myr was calculated from the polarity succession.

confirm that the sedimentary section does not reach the Callovian in this area as was suggested from palynology. A minimum sediment accumulation rate of 170 m/Myr was calculated, reflecting low accommodation space and sediment bypass.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jsames.2019.102235.

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